

1. (20 pts) Calculate the contact maps for the following three proteins (use any program that you wish, or ask me for help):

- a. Concanavalin A (PDB: 3CNA)



- b. Myoglobin (1MBO)



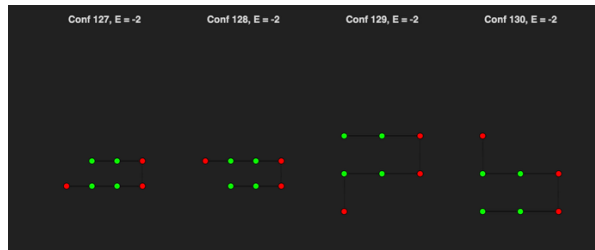
- c. Ubiquitin (1UBQ)



Describe how the secondary structures of these proteins are reflected in the contact matrices.

The hydrogen bonding networks of the secondary structures are represented in the contact matrices. You can clearly see the hydrogen bonding form between parallel beta sheets and vertically through helices.

2. (30 pts) Consider a 7-residue protein with the sequence HHPHHP within the HP-model discussed in lecture. We will explore the conformations of this protein on a 2D-lattice model. I have written a matlab script that enumerates all the possible conformations of this peptide (195 total). The script is available under *Files/Problem_Set_3_HP_model_peptides.m*
 - a. (10 pts) Find the conformation(s) of the folded state, namely the lowest-energy state of the protein, or the state with the most H-H contacts.



- b. (10 pts) Compute the enthalpy and entropy of folding for this protein given that the energy between two non-polar contacts is ϵ .

Since there are 2 H-H contacts, the $\Delta H_{\text{fold}} = -2\epsilon$.

For 195 total combinations and 4 fully folded states,

$$\Delta S_{\text{fold}} = -k_B \ln\left(\frac{195}{4}\right)$$

Therefore, $\Delta S_{\text{fold}} = -3.89 k_B$

- c. (10 pts, extra credit) Compute the contact matrices for all the conformations.

3. (20 pts) Read the D.E. Shaw Science paper, Lindorff-Larsen, Kresten, et al. "How fast-folding proteins fold." Science 334.6055 (2011): 517-520.

Answer the following three questions.

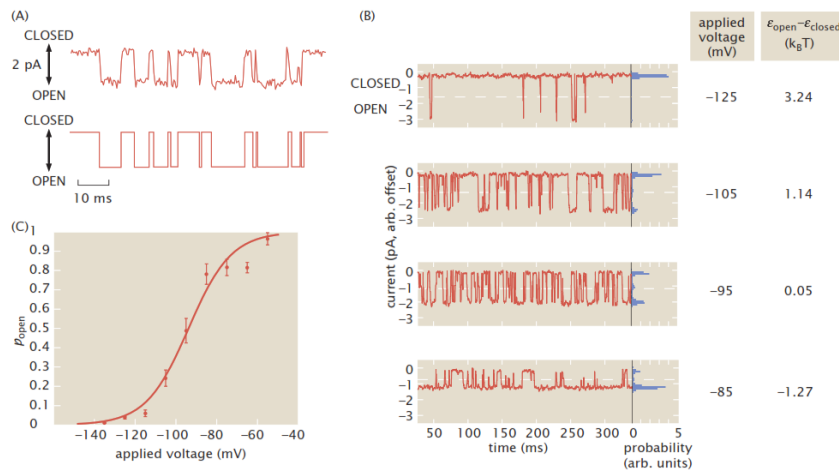
- How does the paper define a folding "pathway"?
They defined a folding pathway as a trajectory for structural elements to form their contacts in a similar order during folding.
- Did the paper find a proteins fold along a variety of pathways, or does a protein generally follow the same well-defined pathway with every folding event?
Proteins often fold along a variety of pathways, however these distinct pathways still share what they described as a "largely common folding nucleus", so the differences in folding pathways are minor differences that they called noise.
- One of the proteins (BBL) was found to undergo folding without going through a "barrier", or transition state. What are the implications of undergoing barrierless folding as postulated by the authors?
They hypothesized that the implications of this could be that the protein has no distinct transition state, folding proceeds smoothly along the energy landscape, or that folding timescales do not separate.

4. (10 pts) Explain in your own words, what the difference is between a transition state and an intermediate, in the context of protein folding.

Transition states in protein folding are highly unstable, dynamic conformations that the protein undergoes to reach its fully folded state. This is the point in which some contacts have formed, but the secondary structure has not yet fully been stabilized due to it being near the top of the free energy landscape.

Intermediates are more stable conformations that the protein may be found in a local minimum going down the energy landscape. This could be a conformation that a protein takes going from a metastable form to a stable form in the energy landscape. These intermediates could hypothetically be isolated.

5. (20 pts) Ion channels conduct ions when they are in the “open” state only. Single-channel conductance measurements help characterize the thermodynamics of gating. Consider the following ion-channel current measurements performed as a function of voltage. The channel is primarily in the closed state at low voltages, and but as voltage increases, the open state becomes more favorable. Based on your knowledge of biophysics, is the gating mechanism a cooperative transition? (support your claim using the measurements below as evidence). Sketch a diagram of what the free energy landscape of the gating process may look like at different voltages.



Voltage gating mechanisms are likely cooperative due to their sensitivity to voltage changes and rapid changes. If the current does not increase gradually but instead rises steeply once a threshold voltage is reached, that supports cooperative gating. This can also be seen in the sigmoidal shape of the graph in (C) as the voltage increases.

